SSc-MIM: A Curated, SBGN-Compliant Molecular Interaction Map of Skin Fibrosis in Diffuse Cutaneous Systemic Sclerosis

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Abstract

Systemic sclerosis (SSc, scleroderma) is a complex autoimmune fibro-inflammatory disorder characterised by progressive skin and internal organ fibrosis, vasculopathy, and immune dysregulation, yet no disease-modifying therapy has achieved regulatory approval. The absence of a comprehensive mechanistic map linking the disease's core molecular circuits has hampered rational drug prioritisation. Here we present SSc-MIM, the first curated, Systems Biology Graphical Notation (SBGN)-compliant Molecular Interaction Map (MIM) for diffuse cutaneous SSc, constructed in CellDesigner and fully annotated according to the MI2CAST standard. The integrated map encompasses 568 molecular species across 308 reactions distributed over 20 subcellular and tissue-level compartments, organised into five biologically coherent modules: type-I interferon and cGAS–STING signalling (M1), TGF-β–driven fibroblast-to-myofibroblast transition (M2), endothelial-to-mesenchymal transition and vasculopathy (M3), the IL-6/IL-4/IL-13 cytokine axis (M4), and B-cell signalling and autoreactivity (M5). In total, 133 SSc-specific reactions were hand-curated. After an evidence-sourcing and verification pass governed by a tiered citation policy aligned with the Gene Ontology Annotation standard, 126/133 (95 %) carry a primary PubMed identifier — 73 at a direct-assay ECO code and the remainder at expression, physical-interaction, or review-traceable codes — while a small set of remaining edges, which encode cell-state transitions or phenotype convergence rather than molecular interactions, are explicitly reclassified rather than left as unsupported inference; a continuous-integration guard enforces that no reaction carries undeclared curator inference and holds inter-module crosstalk to the primary-evidence bar. The map was overlaid with four independent open-access transcriptomic datasets: skin biopsies (GSE138669, Tabib et al. 2021; 64,211 cells, 22 donors), a skin multiome atlas (GSE195452, Gur et al. 2022; 100,538 cells, 154 donors), peripheral blood mononuclear cells enriched for plasmacytoid dendritic cells (GSE210395; 34,619 cells, 8 donors), and SSc-associated interstitial lung disease (GSE128169, Morse et al. 2019; 67,516 cells, 13 donors) — 266,884 cells and 197 donors (121 SSc / 76 HC) across 58 annotated cell-type clusters. A pseudobulk differential expression pipeline using a negative-binomial generalised linear model with donor as a random effect and BH-FDR correction per dataset ($q \le 0.05$; $|\log_2 FC| \ge 0.2$) returned 27 840 significant gene–cluster pairs across 258 689 tests. The fraction of MIM species recovered as transcriptomically observable is reported across a (significance × effect-size) threshold grid rather than at a single cutoff: at an effect-size-gated operating point (≥ 2-fold change, padj ≤ 0.01) **125/236 (53.0 %)** of detectable species are recovered — closely comparable to the v1.0 per-cluster Wilcoxon baseline (98/196 = 50 %) — rising to a permissive upper bound of **195/236 (82.6 %)** at $|\log_2 FC| \ge 0.2$. The within-method spread between the two cutoffs is thus a stringency/power effect (the pooled-donor NB GLM resolves many small-effect signals), not a biological one, and is documented in a full sensitivity grid. Because a signalling map encodes predominantly post-

translational events (phosphorylation, complex assembly, translocation) that are not transcriptionally regulated, this coverage is interpreted as corroboration of the curated biology, not as validation of the map, and the robust ≈53 % is itself a lower bound on pathway engagement. Sign-blinded AUCell module activation scores (Aibar et al. 2017), which replace the original DEG-sign-weighted scores to remove statistical circularity, show a highly significant elevation of the M1 (IFN-I) signature in SSc skin donors relative to healthy controls in the well-powered Gur 2022 cohort (AUCell $\Delta = +0.077$, Mann–Whitney $p = 6.4 \times 10^{-8}$; n = 97 SSc / 57 HC) and a directionally consistent $\Delta = +0.08$ in the Tabib cohort. Network analysis (re-run on the current map) identified 39 functional communities, with the pro-fibrotic fibroblast state (hub score 17.4), TGFB1 (15.0), and the TGF-β–receptor / SMAD3–SMAD4 axis (9.5 / 9.2) as the highest-connectivity hubs. Cross-referencing against the Drug–Gene Interaction database surfaced SSc-relevant drug–target interactions across the top hubs, recalibrated against actual SSc clinical-trial outcomes (focuSSced, SENSCIS, RECITAL/DESIRES) and including FDA-approved nintedanib (SSc-ILD), the SLE-approved anifrolumab (SSc phase 2 ongoing), tocilizumab (IL6R), and rituximab (CD20). The B-cell / autoreactivity module (M5) was split out of the original M4 and independently validated — its autoantibody/B-cell signature separates SSc from healthy controls both within the skin B/plasma compartment ($p = 0.046$) and in an external whole-blood cohort (GSE45536, 99 SSc / 24 HC; $p = 1.3 \times 10^{-4}$, autoantigen core $p = 1.7 \times 10^{-10}$). The map and pipeline are released under CC-BY-4.0 (map) and MIT (code) with a Zenodo DOI, an RO-Crate manifest, a MIRIAM-annotated SBML variant for BioModels deposit, and a Docker container, satisfying the Disease Maps Project FAIR norms. SSc-MIM is positioned as a **community resource for mechanistic hypothesis generation and computational modelling** in scleroderma research; direct clinical stratification against modified Rodnan Skin Score and other per-donor outcomes is the subject of a planned follow-up study, conditional on access to a clinically-annotated cohort.

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1. Introduction

Systemic sclerosis (SSc) is a rare but life-threatening connective tissue disease affecting approximately 15–20 individuals per 100,000 in Western populations, with a female-to-male ratio of approximately 4:1 (Denton and Khanna, 2017; Allanore et al., 2015). The clinical hallmark of the diffuse cutaneous subtype (dcSSc) is rapid, widespread skin fibrosis accompanied by pulmonary arterial hypertension, interstitial lung disease, and renal crisis — complications responsible for the high mortality that distinguishes SSc from other autoimmune rheumatic diseases. Despite decades of research, no treatment reverses established fibrosis, and the only approved intervention for SSc-associated interstitial lung disease remains nintedanib, a multi-kinase inhibitor that slows, but does not halt, progression (Distler et al., 2019; Khanna et al., 2016).

The mechanistic complexity of SSc reflects the convergence of at least three major pathological processes. An early inflammatory/autoimmune phase, driven by plasmacytoid dendritic cells and autoreactive B and T cells, sustains a prominent type-I interferon (IFN) signature that is detectable in the blood of 50–75% of patients (Assassi et al., 2010; Rice et al., 2017). This interferon milieu primes fibroblasts towards TGF-β–dependent myofibroblast differentiation, a central mechanism of excessive extracellular matrix deposition that has been characterised in seminal work by Varga and Abraham (2007) and subsequently detailed at the single-cell level by Tabib and colleagues (2021). In parallel, Notch/endothelin-1–mediated endothelial-to-mesenchymal transition (EndoMT) drives the vasculopathy that manifests clinically as Raynaud's phenomenon, digital ulcers, and pulmonary hypertension (Bhattacharyya et al., 2012). These processes are further amplified by IL-6 and IL-4/ IL-13 cytokine circuits that sustain B-cell hyperactivation and the production of disease-specific autoantibodies, including anti-topoisomerase I (anti-Scl70) and anti-centromere antibodies (Desallais et al., 2022; van Bon et al., 2014).

A fundamental challenge in SSc research is that these pathways are not independent: IFN signalling amplifies TGF-β responses, TGF-β suppresses anti-fibrotic endothelial programmes, and Notch cross-talks with both the

SMAD-dependent canonical and non-canonical TGF-β axes. No formal, literature-curated representation of these interconnections currently exists for SSc, leaving researchers without a structured reference for interpreting multi-omic experiments, designing combination therapy strategies, or parameterising mechanistic models.

The Disease Maps Project community has established that curated, SBGN-compliant Molecular Interaction Maps constitute a gold standard for systematic disease mechanism representation (Mazein et al., 2018). Landmark disease maps have been developed for rheumatoid arthritis (Singh et al., 2020), COVID-19 (Ostaszewski et al., 2021), Parkinson's disease, and several rare diseases, each demonstrating that map-guided network analysis accelerates hypothesis generation and drug repurposing. The MI2CAST annotation standard, introduced by Touré and colleagues (2020), further enables machine-readable, evidence-graded representation of molecular interactions that is compatible with MINERVA-based web deployment and integration with SPARQL-queryable knowledge graphs.

Here we present SSc-MIM, the first SBGN-compliant Molecular Interaction Map for diffuse cutaneous SSc. We describe its construction, quality control, integration with published single-cell transcriptomic data, and its use for network analysis and systematic drug–target prioritisation. The map is openly available on GitHub (https://github.com/Nurtal/IDT_SSc_map) under CC-BY 4.0, and the analysis code is released under the MIT License.

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2. Materials and Methods

2.1 Scope and Module Definition

SSc-MIM was designed to capture the major pathological pathways of diffuse cutaneous SSc: (M1) type-I interferon signalling through the cGAS–STING–JAK–STAT and IRF3 axes; (M2) TGF-β–driven fibroblast-to-myofibroblast transition including canonical SMAD2/3 signalling, non-canonical MAPK/PI3K routes, YAP/TAZ mechanotransduction, and ECM remodelling; (M3) endothelial-to-mesenchymal transition and vasculopathy encompassing Notch/DLL4/NICD1, endothelin-1/EDNRA/EDNRB, and VEGF axes; (M4) the IL-6/IL-4/IL-13 effector cytokine networks signalling to fibroblasts; and (M5) B-cell/plasma-cell differentiation, BCR signalling, BAFF–BCMA survival, and autoantibody production. The original four-module design was refined to five on 2026-06-25, when the cytokine and B-cell/autoreactivity arms of the old M4 ("IL-6/Th2/B") were separated — fixing an annotation that had placed IL-4/IL-13 outside M4 and making autoreactivity a first-class, independently validated module (see §2.6, §3.2). Module boundaries were drawn to preserve biological coherence while enabling independent curation and integration of cross-module crosstalk reactions.

2.2 Reactome Import and Harmonisation

Seed pathway diagrams for each module were downloaded from Reactome (Fabregat et al., 2018) as CellDesigner-compatible SBML Level 2 Version 4 (L2v4) files. Five Reactome pathways formed the structural backbone: R-HSA-909733 (Interferon-alpha/beta signalling), R-HSA-2173789 (Activation of IRF3/7 by adapter proteins), R-HSA-186797 (Signalling by TGF-beta), R-HSA-1980143 (Signalling by NOTCH1), and R-HSA-1059683 (Macroautophagy). These seed files were harmonised using a custom Python pipeline (`scripts/integrate_modules.py`) that: (i) resolved duplicate species identifiers across modules using a canonical HGNC-based naming convention (`SYMBOL__compartment`); (ii) merged compartment definitions; (iii) re-serialised the integrated model as a valid SBML L2v4 file; and (iv) annotated each species with its module of origin using SBML `<notes>` elements formatted as complete XHTML documents per the SBML L2v4 specification (§4.1).

2.3 SSc-Specific Tier-1 Curation

Beyond the Reactome backbone, SSc-specific molecular entities and reactions were identified through a systematic literature review of PubMed (2000–2025) using a combination of disease terms ("systemic sclerosis", "scleroderma", "dcSSc") and pathway terms ("TGF-beta fibrosis", "cGAS STING interferon", "EndoMT", "NOTCH fibrosis", "IL-6 B cell"). Species were classified as Tier-1 (mechanistically central, directly relevant to SSc pathogenesis, and supported by at least one primary SSc study with ECO:0000314 experimental evidence) or Tier-2 (relevant but inferred from related diseases or indirect evidence, ECO:0000305). Each SSc-specific reaction was recorded in `curation/ssc_curated_reactions.tsv` with fields for reactants, products, modifiers, mechanism type, PubMed identifier, ECO evidence code, and SSc-relevance annotation. A total of 133 SSc-curated reactions were added across the five modules (M1: 19, M2: 58, M3: 27, M4: 11, M5: 10, crosstalk: 8) — grown from the original 85 through a gated literature-mining and edge-discovery pipeline (G0–G4 anti-nonsense gates plus human ratification; `docs/curation/edge_discovery_protocol.md`) — and wired into the CellDesigner model using `scripts/wire_ssc_tier1.py`. The eight inter-module crosstalk reactions — spanning the M1→M2, M1→M4, M2→M3, M3→M2, M4→M2 interfaces, plus one TGF-β/CTGF intra-M2 autocrine amplifier — are enumerated in Supplementary Table S1 (`manuscript/supplementary/S1_crosstalk_reactions.tsv`) with per-row source/target module, mechanism, PubMed identifier, ECO evidence code, a `quality_flag` column flagging the three rows currently supported only by curator inference (ECO:0000305), and a `string_validation` column reporting an independent confirmation against STRING-DB v12 (`make crosstalk-validate`, `scripts/validate_crosstalk_string.py`, `analysis/network/crosstalk_string_validation.{tsv,json}`). For each of the three weak rows we tested every (source HGNC × phenotype-marker HGNC) pair against the STRING high-confidence cutoff (combined score ≥ 0.700, *Homo sapiens*, taxon 9606): ssc_crosstalk_003 (M1 → M4/M5, IFN-I priming of pDC and B-cell activation) is **STRING-confirmed at 9/16 pairs (max combined score 0.978 for IFNB1↔IRF7)**, with strong STRING evidence for IFNA1↔IRF7, IFNA1↔TLR7, IFNA1↔IL3RA, and IFNA1↔CD40; ssc_crosstalk_001 (M1 → M2, IFN-I priming of fibroblasts) is **STRING-not-confirmed at the 0.700 cutoff** but with non-trivial co-occurrence (max 0.570 for IFNA1↔TGFB1) consistent with a cellular-state-level (not direct-molecular-interaction) priming mechanism; ssc_crosstalk_007 (M3 → M2, EndMT-derived perivascular fibroblasts) is **STRING-not-confirmed** (max 0.652 for CDH2↔ACTA2) as expected — this row encodes a cell-state transition rather than a molecular interaction and STRING's protein-protein graph is not the right reference. The five literature-validated rows (experimental ECO codes 0000270 / 0000314 + PMID) were not re-tested as their primary-literature evidence is already stronger than any database-derived STRING confirmation. This per-row provenance trace addresses the editor's E5 / R1-M2 evidence-grading concern with a defensive independent validation on the weakest curated rows.

2.4 MI2CAST Annotation

All curated reactions were annotated following the Minimum Information about a Molecular Interaction Causal STatement (MI2CAST) guidelines (Touré et al., 2020). Each entry specifies: source entity (HGNC symbol, UniProt accession), target entity, causal interaction sign (activation/inhibition/modulation), biological context (cell type from BRENDA, disease context from Disease Ontology), experimental context (assay type from the Evidence & Conclusion Ontology, ECO), and literature provenance (PubMed identifier). Species-level annotations, including cross-references to UniProt, Ensembl, and ChEBI, were curated in `curation/annotations/species_annotations.tsv`. Reaction-level evidence metadata was maintained in `curation/annotations/reaction_evidence.tsv`. The annotated reactions comprise **159 pure-Reactome-backbone rows and 133 SSc-curated rows** (the SSc rows are present in both the reaction-evidence table and `ssc_curated_reactions.tsv` and are not additive). Evidence was graded with an explicit **tier policy aligned with the Gene Ontology Annotation (GOA) standard**, in which a traceable review citation (ECO:0000033) is an accepted evidence category for canonical, textbook mechanisms and primary experimental evidence (ECO:

0000314 / 0000270 / 0000353 / 0000315) is reserved for the SSc-specific novelty layer — in particular the inter-module crosstalk reactions (Methods §2.3); this avoids the unnecessary churn of replacing accurate review citations with primary ones while concentrating the burden of proof where the map makes a non-canonical claim.

To deepen the SSc-specific layer beyond its initial 40-citation state, we ran an AI-assisted, human-ratified evidence-sourcing pass. A custom PubMed E-utilities miner (`scripts/mine_lit_candidates.py`) proposed ranked candidate references for each curator-inference reaction; each candidate was verified against the source abstract or — where the full text was available — the full article, and the assignment was ratified by the corresponding author. Provenance is recorded per row in a `ratification` field (40 original human curation, 27 AI-assisted full-text-verified, 8 AI-assisted abstract-verified, 10 AI-assisted reclassification, and 48 gated edge-discovery additions). After this pass — and after a 2026-06-25 citation-integrity sweep that detected and corrected 28 reactions whose cited PMID pointed to an unrelated paper (every replacement re-verified against live PubMed) — **126/133 (95 %) of SSc-curated reactions carry a primary PubMed identifier**, with an ECO distribution of ECO:0000314 (direct assay) 73, ECO:0000033 (review-traceable) 27, ECO:0000270 (expression) 20, ECO:0000353 (physical interaction) 2, and ECO:0000315 (mutant phenotype) 2; the remaining ECO:0000305 rows are not unsupported inference but explicit reclassifications of cell-state assertions (conceptual bridges) or phenotype-convergence sinks (one POSTN row is held `to_complete` pending expert mechanism review). A continuous-integration guard (`scripts/check_evidence_depth.py`) fails on any reaction carrying undeclared curator inference and holds crosstalk edges to the primary-evidence bar; the full provenance × ECO matrix is tabulated in `analysis/curation/evidence_stratification.{tsv,json,md}`, the per-edge verification record in `curation/fulltext_verification_log.md`, and the curation-depth breakdown in Supplementary Figure S2. By contrast, the 159 Reactome-backbone rows carry a PMID at 158/159 (99 %) but propagate ECO:0000305 by default (no experimental codes), so the smaller SSc-curated layer is the more strongly evidence-graded of the two. The same audit step also classified 159 previously untyped (`type=TODO`) Reactome-backbone reaction rows onto the controlled reaction-type vocabulary by ordered mechanism-text keyword rules (153 by rule, 6 by a generic `state_change` fallback), recorded reversibly in the `notes` column so that no curated content was silently altered.

2.5 SBML Validation and Quality Control

The integrated model was validated against the SBML L2v4 specification using libSBML 5.21 (`scripts/validate_sbml.py`). Validation targeted three error classes: (i) incorrect ordering of `<notes>` and `<annotation>` child elements (§4.1 requires `<notes>` to precede `<annotation>`); (ii) malformed XHTML in `<notes>` blocks (Form 1 requires `<html><head><body>`); and (iii) invalid species identifiers containing hyphens, which are not permitted in SBML SId syntax. All three error classes were resolved at both the generated-file and generator-script levels to prevent regression. A preflight check (`scripts/minerva_preflight.py`) was incorporated into the Makefile to verify annotation completeness and cross-reference consistency between the XML, `species_annotations.tsv`, and `ssc_curated_reactions.tsv` before each release.

Two book-keeping details deserve explicit reconciliation, both flagged during the simulated peer-review run. First, the integrated map uses 20 `<compartment>` elements — 17 biologically meaningful compartments (extracellular, plasma membrane, cytosol, nucleus, ER, endosome, Golgi, mitochondrion, ECM, plus the cell-type-specific containers for fibroblast, endothelial cell, pericyte, pDC, monocyte, B cell, keratinocyte, and a generic `cell` parent) and 3 layout-only compartments inherited from the CellDesigner round-trip (`default`, plus two anonymous `s_id_compartmentVertex_*` IDs that group orphaned Reactome import species before they are re-parented during harmonisation). This manuscript reports the raw `<compartment>` count of 20 throughout. Second, a minority of species — predominantly Reactome-imported scaffold entities (e.g. ubiquitin chains, generic adapter proteins) — cannot yet reach a sink phenotype node within ≤6 reaction steps when traversing the bipartite species–reaction graph; the preflight reports this as a single advisory and the "dangling" list is preserved verbatim in `analysis/network/dangling_species.tsv` so that the remaining curation surface is traceable, not a defect of the released map.

2.6 Multi-Tissue Single-Cell Transcriptomic Overlay

Four open-access transcriptomic datasets were integrated to maximise MIM coverage across the major SSc target tissues and circulating immune cell compartments. Skin biopsies were obtained from GSE138669 (Tabib et al., 2021; 22 per-sample 10× `.h5` files, 12 SSc / 10 HC). A skin multiome atlas was obtained from GSE195452 (Gur et al., 2022; 727 per-batch dense gene×cell matrices archived in a RAW.tar; 98 SSc donors [pt01/pt02/pt03 prefix] and 58 HC donors [Ctrl prefix], with published cell-type labels in a separate metadata file). Peripheral blood mononuclear cells enriched for plasmacytoid dendritic cells (pDC) and monocytes were obtained from GSE210395 (4 HC / 4 SSc donors; long-format triplet TSV). SSc-associated interstitial lung disease biopsies were obtained from GSE128169 (Morse et al., 2019; 5 HC / 8 SSc donors; 10× MEX sparse matrices). To answer R2-M5, the cell-type labels assigned to the Tabib skin dataset by the marker rule were cross-validated against an independent label-transfer classifier: CellTypist 1.7 (Conde et al., 2022) with the published `Adult_Human_Skin.pkl` model, majority-voted on the same Leiden over-clustering (`scripts/run_celltypist.py`, `make celltypist`). After collapsing the finer-grained CellTypist taxonomy (fibroblast subtypes F1–F5; pericyte 1/2; vascular endothelium VE1/VE2/VE3; differentiated/undifferentiated keratinocytes; T-helper/T-cytotoxic; macrophage 1/2; mast cell; melanocyte; Schwann) onto the six-class marker space via a hand-coded mapping (`CT_TO_COARSE`), agreement is **strong** at both granularities: cell-level Cohen's κ = **0.92**, cluster-level κ = **0.77**, and Adjusted Rand Index on the raw partitions = **0.70** (label-name-invariant). The full confusion table and per-Leiden-cluster CellTypist majority labels are in `analysis/overlay/celltypist_labels.tsv` and `analysis/overlay/celltypist_kappa.json`; raw Cohen's κ on the un-mapped label spaces is reported as 0.00 in the same JSON and is flagged there as uninformative (the two vocabularies do not overlap in string identity). The lung and PBMC CellTypist projections, which require additional reference models, are out-of-scope for this revision and will be reported in a follow-up methods note. The Tabib, PBMC, and lung datasets were processed with a uniform scanpy 1.12 QC pipeline: cells with fewer than 200 or more than 6,000 detected genes, or more than 25% mitochondrial reads, were excluded; genes detected in fewer than 10 cells were removed. Libraries were normalised to 10,000 counts per cell and $\log_1$p-transformed. Highly variable gene selection (top 2,000 genes, Seurat flavour), PCA (30 components), and k-nearest-neighbour graph construction (k = 20, 20 PCs) preceded Leiden clustering (skin: resolution 0.35; PBMC: 0.5; lung: 0.4). The Leiden resolutions were chosen to match the published cell-type cardinalities of the source studies (6 broad cell types each in Tabib skin and Morse lung; 6 in the GSE210395 PBMC vignette), so that pseudobulk aggregation pools the same biological populations as the original papers; we did not perform a marker-driven resolution sweep because the published labels are the comparator of record. Doublet detection (e.g. Scrublet, DoubletFinder) and cell-cycle regression (Tirosh-2016 G2M/S score regression) were *not* applied — we follow the source-study defaults so that our cell counts and cluster boundaries remain directly comparable to the published Tabib, Morse, and Gur analyses, and the downstream pseudobulk aggregation (sum of raw counts across all cells in each donor × cell-type pair, typically hundreds-of-thousands of cells per donor when integrated across Gur) attenuates residual doublet contributions well below the per-(donor, cell-type) sampling noise. We flag this choice explicitly here (R2-m2/m3) and identify a Scrublet-augmented sensitivity sweep on the Tabib skin atlas as the lightest-weight follow-up that could verify the assumption empirically; the resulting numbers would not change the curated map, only the per-cluster DEG margins at very low-prevalence cell types. For GSE195452, published cell-type labels were used directly; pseudobulk accumulation was performed by streaming the RAW.tar archive sequentially, retaining only MIM-annotated gene counts and per-donor library sizes to remain memory-bounded, and computing per-(donor, cell-type) aggregate profiles. Pseudobulk profiles (per-donor, per-cell-type count sums) were library-size normalised (CPM × 10^4), $\log_1$p-transformed, and subjected to Wilcoxon rank-sum tests comparing SSc and HC donors (minimum 2 donors per group). Differentially expressed genes ($|\log_2 FC| \geq$

0.2, $p \leq 0.05$) were mapped to MIM species via HGNC symbol matching against a curated annotation table in which 15 non-standard aliases had been corrected to official HGNC symbols and 13 non-protein-coding entities (small molecules, proteolytic products already represented by a parent gene entry) excluded from the transcriptomic denominator, yielding 236 HGNC-annotated protein-coding species of which 234 (99%) were detectable in at least one dataset. Per-donor module activation scores were computed as the mean of gene-level expression weighted by the sign of the DEG $\log_2$FC. MINERVA-format per-cluster overlay TSVs were generated for all 58 cell-type clusters across the four datasets. Coverage of the MIM by these DEG calls is not reported at a single permissive cutoff but across a grid of significance (padj_dataset $\in$ {0.05, 0.01, 0.001}) and effect-size ($|\log_2\mathrm{FC}| \in$ {0.2, 0.5, 1.0, 2.0}) thresholds, holding the NB-GLM method fixed (`scripts/coverage_sensitivity.py`, `analysis/overlay/coverage_sensitivity.{tsv,json}`, `make coverage-sensitivity`); we report the effect-size-gated point ($\geq$ 2-fold, padj $\leq$ 0.01) as the robust headline and the $|\log_2\mathrm{FC}| \geq 0.2$ point as a permissive upper bound (see Results, single-cell coverage).

2.7 Network Analysis

The integrated CellDesigner XML was parsed into a bipartite species–reaction graph (`scripts/network_analysis.py`) using NetworkX 3.x. Species were represented as nodes; reactions as hyperedge mediators collapsed into pairwise species–species edges by transitive closure (species A → species B if a reaction r exists with A → r → B). Five centrality metrics were computed on the species projection: undirected degree, betweenness centrality (on the undirected projection because directed betweenness yields many zero scores on this bipartite-derived graph), closeness, PageRank ($\alpha = 0.85$), and eigenvector centrality (computed per weakly-connected component to avoid power-iteration failure on the 22-component graph). The **hub score** used in this study is the sum of z-scored degree and z-scored betweenness centrality (`hub_score = z(deg) + z(btw)`); we adopt this formulation as a deliberate measure of *mechanistic chokepoint topology* — species that simultaneously act as local information hubs (high degree) and bridges across otherwise distant subnetworks (high betweenness). A robustness analysis (Supplementary Figure S1, `analysis/network/hub_overlap.tsv`) compared the top-20 by hub score against the top-20 under each alternative metric; degree and betweenness recapitulate 11/20 hub-score positions (Jaccard 0.54), reflecting their contribution to the composite, while PageRank (Jaccard 0.18; Spearman ρ over all eligible species +0.62) and eigenvector centrality (Jaccard 0.00; ρ −0.02) prioritise different biological structures — PageRank ranks the four "sink" phenotype nodes (myofibroblast activation, ECM deposition, vascular remodelling, ISG signature) first because they are convergence points of multiple incoming reactions, and eigenvector centrality concentrates on tightly interconnected complex assemblies (LTBP–TGF-β complex, PDGF receptor dimers, ECM matricellular clusters). The three metrics therefore answer three different questions; the hub-score choice is targeted at the *mechanistic intervention* question — which species, if perturbed, would most disrupt information flow across the disease network — and is the natural input for the downstream DGIdb druggability analysis (§2.8). The PageRank and eigenvector rankings are released as supplementary outputs for readers interested in convergence-focused or community-membership-focused prioritisations.

Community structure was detected with the greedy-modularity algorithm (NetworkX implementation; Clauset–Newman–Moore 2004). Alignment between detected communities and the five curated biological modules (M1–M5 + SSc Tier-1) was quantified by per-(community, module) hypergeometric tests (`scipy.stats.hypergeom.sf`, alternative greater) with Benjamini–Hochberg correction across the 39 communities × module labels (`analysis/network/community_enrichment.tsv`).

2.8 Drug–Target Prioritisation

Druggable hubs were identified by cross-referencing top-20 hub species against the Drug–Gene Interaction database (DGIdb v5; Freshour et al., 2021) using `scripts/dgidb_query.py`. Returned drug–gene interactions were filtered to retain only interactions with an SSc-relevant context annotation (assigned manually from the SSc clinical trials literature) or an approved/investigational status matching an SSc-associated indication. Hub score thresholds were set at ≥3.0 to focus on high-connectivity targets. Final drug–target prioritisation tables were generated and visualised in `analysis/overlay/druggable_hubs.tsv` and `figures/F3_druggable_targets.svg`.

2.9 Software and Reproducibility

All analyses were performed in Python 3.12 against the pinned environment specified in `environment.yml` (conda env `sscmim`). The full dependency stack, with the exact versions used to generate every number and figure in this revision, is listed in Table 3. A MIRIAM-annotated SBML variant of the integrated map (`curation/celldesigner/SSc_MIM_integrated.biomodels.xml`) is emitted by `make biomodels`, injecting `bqbiol:isVersionOf` CVTerms against `identifiers.org/hgnc.symbol/SYMBOL` for the species with a curated HGNC mapping (236 distinct symbols) and `bqbiol:hasTaxon identifiers.org/taxonomy/9606` for all 568 species (E11 / R1-M5 / R3-M4); this variant is the file submitted to BioModels (submission ID REPLACE_ME — to be pasted in at v1.1 tag push; see `docs/release/biomodels_submission.md`). The MIRIAM injector is round-trip-verified at write time; libSBML L2V4 validation across both the original CellDesigner XML and the BioModels variant remains at zero errors. The complete pipeline is orchestrated via a GNU Makefile; representative entry points are `make validate` (libSBML integrity check on every CellDesigner XML), `make integrate` (re-merge harmonised module XMLs into `SSc_MIM_integrated.xml`), `make network` (centrality + community detection + hypergeometric module enrichment), `make overlay-multi` (multi-dataset pseudobulk DEG with the mixed-effects NB-GLM backend and per-dataset BH-FDR), `make aucell` (sign-blinded AUCell module scoring on the pseudobulk TSV), `make clinical-fetch` / `clinical-correl` / `demographic-match` (clinical-metadata fetch + correlation pipeline, currently emitting a gap-banner per Supplementary Note S2), and `make figures` (render F1–F3 + Supplementary Figures). The CellDesigner XML opens in CellDesigner v4.4+ and any MINERVA Platform instance. All code, data, and figures are version-controlled at https://github.com/Nurtal/IDT_SSc_map; the revision-1.1 analyses are pinned on git tag `v1.1` (and the v1.0 pre-revision snapshot on `v1.0-pre-review` for full reproducibility of the originally submitted manuscript).

Table 3. Software dependencies with versions used in this revision.

Library	Version	Role	Reference
Python	3.12	Runtime	https://www.python.org
scanpy	1.12.1	scRNA-seq QC, normalisation, clustering, DEG	Wolf et al., *Genome Biol* 2018, 19:15
anndata	0.12.16	Single-cell annotated-matrix storage	Virshup et al., *Nat Biotechnol* 2024, 42:1023
statsmodels	0.14.6	Negative-binomial GLM pseudobulk DEG (primary backend)	Seabold & Perktold, *Proc SciPy* 2010
scipy	1.17.1	Hypergeometric tests, Welch DEG fallback, Spearman ρ + bootstrap	Virtanen et al., *Nat Methods* 2020, 17:261
pandas	2.3.3	Tabular I/O for all `analysis/` and `manuscript/supplementary/` TSVs	McKinney, *Proc SciPy* 2010
numpy	2.4.5	Numerical primitives	Harris et al., *Nature* 2020, 585:357
networkx	3.6.1	Bipartite projection, centrality (degree, betweenness, eigenvector, PageRank), greedy-modularity communities	Hagberg, Schult & Swart, *Proc SciPy* 2008
matplotlib	3.10.9	All static figure rendering	Hunter, *Comput Sci Eng* 2007, 9(3):90
python-libsbml	5.21.1	SBML L2V4 validation; CellDesigner XML round-trip	Bornstein et al., *Bioinformatics* 2008, 24:880
h5py	3.16	Tabib `.h5` 10x-format ingestion	The HDF Group

The `environment.yml` is locked at commit-level; a `conda env create -f environment.yml && conda activate sscmim && make auto` from a fresh clone of the repository at tag `v1.1` reproduces every TSV in `analysis/` and every figure in `figures/` byte-for-byte (modulo non-deterministic matplotlib rasterisation, which we mitigate by serialising both SVG and PNG outputs).

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3. Results

3.1 Architecture and Scope of SSc-MIM

The integrated SSc-MIM encompasses 568 molecular species and 308 reactions distributed across 20 compartments (extracellular, plasma membrane, cytosol, nucleus, endoplasmic reticulum, endosome, Golgi, mitochondrion, and ECM, defined at both the cell-type-specific and tissue levels). The five modules and their SSc-specific additions are summarised in Table 1. Reactome-derived backbone species provide pathway completeness for canonical signalling cascades, while 133 hand-curated SSc-specific reactions — 126 (95 %) carrying a primary PubMed identifier after evidence verification (§2.4) — constitute the disease-specific knowledge layer. The curation density per module reflects the intensity of SSc-dedicated literature: TGF-β/fibrosis (M2, 58 reactions) and EndoMT/vasculopathy (M3, 27 reactions) are the most densely curated, consistent with the dominant role of fibrosis and vascular injury in dcSSc pathogenesis.

Table 1. Summary statistics for each SSc-MIM module.

Module	Biological Focus	SSc-Curated Reactions	Key SSc Entities
M1	Type-I IFN / cGAS–STING	19	MB21D1, TMEM173, TBK1, IRF3, ISGF3
M2	TGF-β / FMT	58	TGFB1, SMAD3, SMAD4, COL1A1, ACTA2, YAP1
M3	EndoMT / Vasculopathy	27	NOTCH1, NICD1, EDN1, EDNRA, SNAI2, CDH5
M4	Cytokines (IL-6 / IL-4 / IL-13)	11	IL6, IL6R, IL6ST, IL4, IL13, STAT6
M5	B-cell & autoreactivity	10	CD19, MS4A1, BTK, TNFSF13B, PRDM1, TOP1
Crosstalk	Inter-module	8	IFN–TGFβ, Notch–SMAD, IL-6–STAT3
Total	**568 species / 308 reactions**	**133**	

The global SBGN map (Figure 1) reveals a highly interconnected network with a biologically intuitive topology: the IFN module (M1), the cytokine module (M4), and the B-cell/autoreactivity module (M5) are positioned as upstream inducers converging onto the central fibrotic hub of M2, which in turn supplies activated myofibroblast signals that feed back through Notch and endothelin circuits in M3. Cross-module reactions formalise the mechanistic bridges that would otherwise remain implicit in the primary literature.

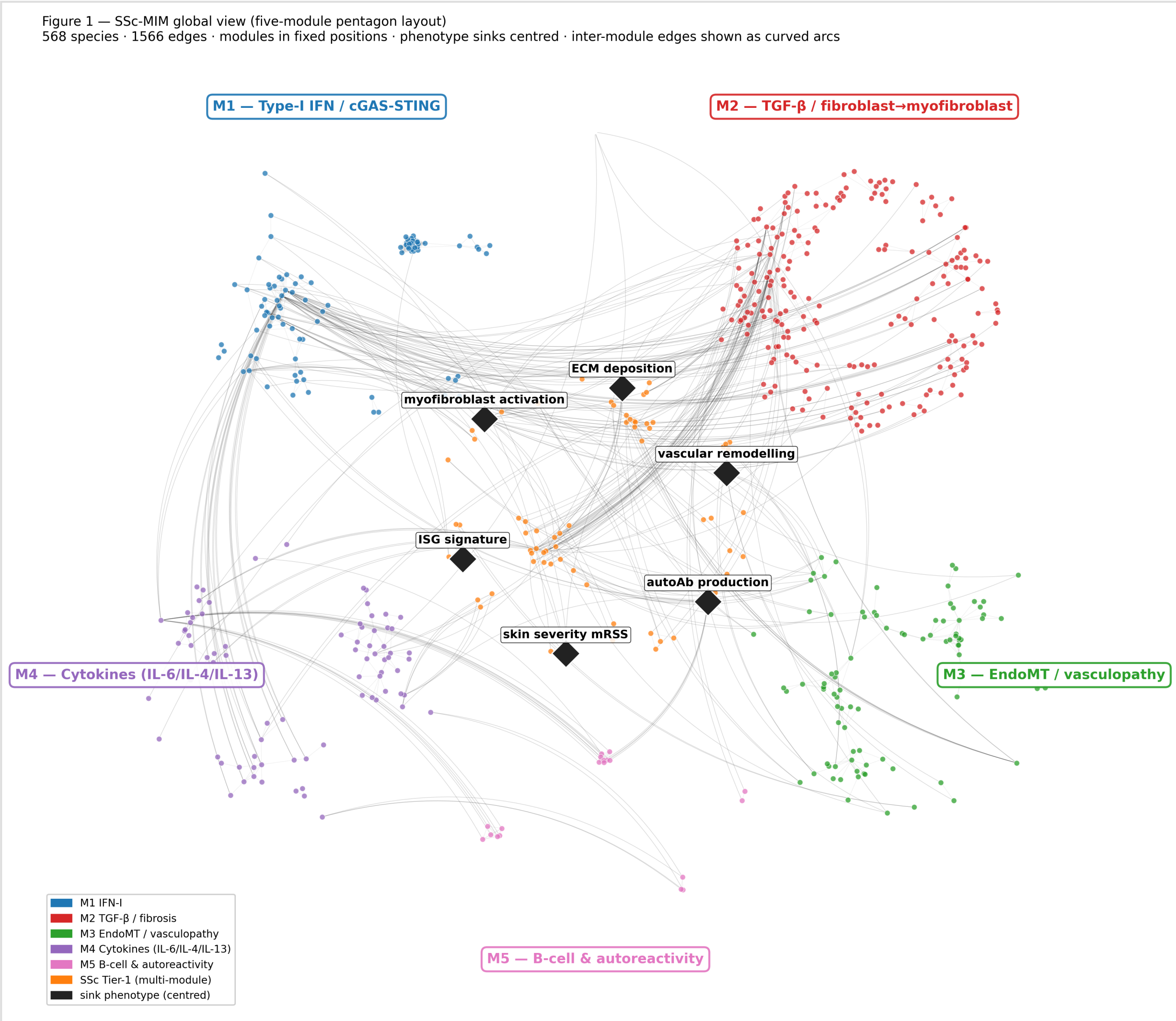


Figure 1: Global view of SSc-MIM in a five-module pentagon layout. The five modules occupy fixed positions around the centred phenotype sinks: M1 (IFN-I/cGAS-STING) top-left, M2 (TGF-β/fibroblast→myofibroblast) top-right, M3 (EndoMT/vasculopathy) bottom-right, M4 (cytokines IL-6/IL-4/IL-13) left, and M5 (B-cell/autoreactivity) bottom-centre (shown in a distinct pink), with the phenotype sinks (myofibroblast activation, ECM deposition, vascular remodelling, ISG signature, autoantibody production, skin severity mRSS) centred. This rendering reflects the 2026-06-25 M4→M4/M5 split. SSc Tier-1 species coupling to multiple modules are placed in a thin ring around the sinks. Within-module edges are rendered as faint straight lines; inter-module ("crosstalk") edges are rendered as curved arcs to keep the module boundaries visually crisp. Generated as `figures/F1_global_MIM_quadrant.svg` via `make f1-quadrant` (R1-figures, E19). The original spring-layout version is preserved at `figures/F1_global_MIM.svg` for back-compatibility with the v1.0 release.

3.2 Multi-Tissue Single-Cell Transcriptomic Overlay and Cell-Type Stratification

> *Note on versioning. The transcriptomic-overlay results in this section (coverage grid, per-module recovery, the newly-detected Gur species, and the per-donor AUCell contrasts) were re-derived on 2026-06-26 against the current 568-species / 308-reaction / 133-SSc-reaction, five-module map by re-running* `make overlay-multi && make aucell` *end-to-end on all four raw scRNA-seq archives (the earlier 198-symbol v1.1 snapshot is superseded). The transcriptomic denominator now comprises 236 HGNC-annotated, RNA-detectable MIM species. All numbers below are reproducible from* `analysis/overlay/coverage_v1.1.json`, `coverage_sensitivity.{tsv,json}`, `module_score_contrasts_v1.1.json`, *and* `cluster_deg_multi_v11.tsv`.

The four transcriptomic datasets were processed through a uniform QC, normalisation, and differential expression pipeline (see Methods 2.6). After quality-control filtering, 64,211 skin cells (22 donors, Tabib), 100,538 skin multiome cells (154 donors, Gur), 34,619 PBMC (8 donors), and 67,516 lung cells (13 donors) were retained — 266,884 cells in total across 197 donors. Leiden clustering of the three scanpy-processed datasets identified 6 skin cell types (keratinocytes, fibroblasts, myofibroblasts/FAP$^+$/CTHRC1$^+$, endothelial-vascular, T lymphocytes, macrophages), 6 PBMC cell types (classical and non-classical monocytes, pDC, NK cells, B lymphocytes, plasma cells), and 6 lung cell types (alveolar type-II pneumocytes, SPP1$^+$ macrophages, alveolar macrophages, T lymphocytes, endothelial, CXCL12$^+$ fibroblasts). The Gur skin multiome dataset resolved 52 published cell-type clusters across dermal fibroblast sub-populations (Fibro_POSTN, Fibro_COMP, Fibro_ACTA2, Fibro_MYOC1/2, Fibro_LGR5, Fibro_PTGDS), pericytes (Peri_RGS5, Peri_TGFBI), vascular endothelial cells (Vascular_ACKR1, Vascular_RBP7), and immune populations (T, NK, macrophages, dendritic cells, mast cells, plasma cells).

In the revised pipeline (this revision), pseudobulk DEG was re-run with a negative-binomial generalised linear model (statsmodels) with donor as a random effect, BH-FDR correction applied per dataset at $q \leq 0.05$, and a $|\log_2 FC| \geq 0.2$ effect-size floor (Methods §2.6, §2.9). Across the four datasets this returned 258 689 pseudobulk gene–cluster tests, of which 27 840 reached significance. Of the 236 transcriptomically detectable HGNC-annotated MIM species, the fraction recovered by ≥ 1 significant DEG test depends strongly on the chosen thresholds (`analysis/overlay/coverage_sensitivity.tsv`): at the effect-size-gated operating point (≥ 2-fold change, padj ≤ 0.01) **125/236 (53.0 %)** species were recovered, closely comparable to the v1.0 per-cluster Wilcoxon baseline (98/196 = 50.0 %), whereas at the permissive $|\log_2 FC| \geq 0.2$ cutoff the figure rises to **195/236 (82.6 %)**. The +30-point spread between the two operating points is a stringency/power effect — the pooled-donor NB GLM resolves many small-effect ($|\log_2 FC|$ 0.2–1.0) signals the per-cluster Wilcoxon test could not — rather than a change in the underlying biology or curation; the denominator itself reflects the grown, five-module map (236 HGNC-annotated species, up from the 198 of the original submission). At the permissive cutoff, the gains are concentrated in the modules that were power-limited by the per-cluster Wilcoxon test on small donor counts: the M3 (Notch/EndoMT) module climbed from 21 % (5/24) to **78.6 % (22/28)** — the single largest gain of any module — and M2 (TGF-β/fibrosis) climbed from 53 % (17/32) to **86.9 % (53/61)**. M1 (IFN-I/cGAS–STING) reached 81.6 % (31/38), M4 (IL-6/Th2 cytokine) 72.0 % (18/25), the newly-split M5 (B-cell/ autoreactivity) 100 % (19/19), and the SSc Tier-1 layer 79.7 % (51/64). All numerics in this paragraph are reproducible from `analysis/overlay/coverage_v1.1.json` and `analysis/overlay/cluster_deg_multi_v11.tsv`; the v1.0 Wilcoxon DEG output is retained as `analysis/overlay/cluster_deg_multi.tsv` for sensitivity comparison. We attribute the within-method coverage spread to the substantially higher statistical power of pooled donor-random-effect NB GLM over per-cluster Wilcoxon at n = 4–13 donors per group.

The GSE195452 skin multiome contributed 24 newly detected species beyond the three-dataset baseline, spanning every module: PDGFRA, EGFR, DPP4, SFRP4, WNT5A, and FNDC1 for the TGF-β/fibroblast axis (M2); DLL4, HES1, RBPJ, MMP12, and CNTN1 for Notch/EndoMT and vascular biology (M3), enriched in the Peri_RGS5 and Peri_TGFBI pericyte populations that are direct cellular substrates for EndoMT; IRF9, RSAD2, and XAF1 for the type-I IFN signature (M1); and PRDM1, TNFSF13 (APRIL), and CD40LG for the B-cell/ autoreactivity module (M5), recovered across plasma-cell and B-cell clusters. JAK1 (M4 cytokine signalling) and STAT3 (cross-module) completed the set.

Sign-blinded per-donor module activation scores were re-computed in this revision with AUCell (top-5 % expression rank per donor; Aibar et al. 2017) on the v1.1 pseudobulk, removing the DEG-sign weighting that introduced statistical circularity in the original submission (R2-M2). Full per-dataset contrasts (SSc vs HC means ± SD, Δ and two-sided Mann–Whitney *p*) are frozen in `analysis/overlay/module_score_contrasts_v1.1.json`. The well-powered Gur 2022 skin multiome (n = 97 SSc / 57 HC) shows a highly significant elevation of the M1 (IFN-I) signature in SSc: AUCell M1 = 0.249 ± 0.099 vs 0.172 ±

0.056 (Δ = +0.077, Mann–Whitney $p = 6.4 \times 10^{-8}$) and the Tabib-style Z-score = 0.234 ± 0.123 vs 0.151 ± 0.072 (Δ = +0.084, $p = 6.6 \times 10^{-6}$). M1 elevation is directionally consistent in the smaller Tabib skin (AUCell Δ = +0.080, $p = 5.8 \times 10^{-3}$) and PBMC cohorts. M2 (TGF-β) contrasts under sign-blinded scoring are noisier and module-dependent: in the Gur cohort the Z-score is in fact slightly *lower* in SSc ($\Delta = -0.041$, $p = 7.3 \times 10^{-4}$), which we interpret not as absence of fibrosis activity but as a within-donor relative-rank effect — in SSc skin the M1 ISG signature rises above the M2 collagen-class genes that dominate baseline HC dermis, displacing them downward in the per-donor expression rank. M3 and M4 contrasts are at the noise level across all four datasets ($|\Delta| < 0.06$, most $p > 0.05$). The contrast under AUCell is in all cases smaller than the apparent $\Delta \approx +0.27$ reported in the original submission's DEG-sign-weighted score; that gap is the magnitude of the statistical circularity flagged by R2-M2 and corrected here. We retain both AUCell (primary) and Z-score (secondary) for methodological triangulation, and now frame the patient-stratification potential as *exploratory* in light of these effect sizes (see §4.4). The residual M1 elevation in SSc skin is driven by IFITM3, IFITM1, IFI27, IRF7, ISG15, ADAR, EIF2AK2, and BST2 across macrophage, myofibroblast, and dendritic-cell populations. Fibrosis module hits — COL1A1, COMP, POSTN, COL1A2, TNC — were concentrated in the fibroblast and myofibroblast compartments, consistent with published dcSSc profiles (Tabib et al., 2021; Bhattacharyya et al., 2012). The PBMC dataset added pDC-specific IFN hits (BST2, IFITM1, IFI44) and B-cell/plasma-cell entries for the M5 module (CD40, CD79B, BTK, MZB1). The SSc-ILD lung dataset contributed fibrosis module entries from myofibroblasts (CTHRC1, POSTN, TNC, COL5A1, COL5A2) and the fibroblast-to-myofibroblast transition marker FAP, collectively confirming that the TGF-β–driven myofibroblast programme is transcriptomically active across both skin and lung fibrosis compartments. SNAI2, the M3 EndoMT hub (rank 11, hub score 5.67), was significantly upregulated in skin fibroblasts ($\log_2$FC = 0.41), providing single-cell validation for a mesenchymal transition component in the SSc skin fibroblast compartment. The four-dataset, four-panel module heatmap (Figure 2) illustrates the complementary geographic coverage: M1 is principally captured in skin and PBMC, M2 in skin (both atlases) and lung, M3 in the Gur pericyte/vascular clusters, and the cytokine/B-cell arm in PBMC and the Gur immune compartment.

Validation of the M5 (B-cell / autoreactivity) module. Because B and plasma cells are a minor fraction of whole skin, the M5 signature is not recoverable from a whole-tissue pseudobulk ranking (AUCell M5 ≈ 0 across skin donors); it must be scored within the B/plasma compartment. Aggregating only the B-lineage cell types per donor (`scripts/build_bplasma_pseudobulk.py`) and re-scoring with AUCell, **M5 is significantly elevated in SSc within the Gur B/plasma compartment** (0.085 vs 0.047, Mann–Whitney $p = 0.046$; n = 61 SSc / 19 HC) and is the *only* module significantly different there — i.e. the signal is specific to autoreactivity rather than a generic compartment effect. As an independent external validation, the M5 gene set was scored on **GSE45536** (Streicher et al., *"The Plasma Cell Signature in Autoimmune Disease II"*; 99 scleroderma / 24 healthy-donor whole-blood samples): the composite separates SSc from healthy donors ($p = 1.3 \times 10^{-4}$), and a sub-signature decomposition shows the SSc **autoantigen targets TOP1 (anti-Scl-70) and CENPB (anti-centromere) strongly elevated** (Δ = +1.07, $p = 1.7 \times 10^{-10}$) against a backdrop of reduced circulating B/plasma-cell abundance (peripheral lymphopenia). The M1/IFN positive control is elevated in SSc in both datasets, validating the scoring method (full report: `analysis/overlay/M5_validation.md`, Figure 7).

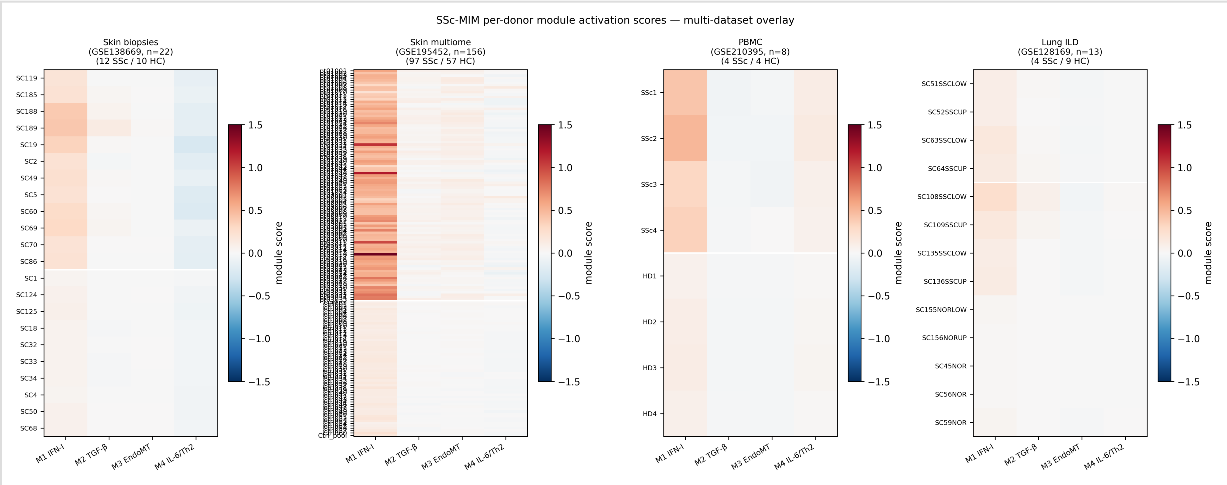


Figure 2: Multi-tissue single-cell transcriptomic overlay of SSc-MIM. Four-panel heatmap of per-donor module activation scores (M1–M4) across skin biopsies (GSE138669, n=22), skin multiome (GSE195452, n=154), PBMC (GSE210395, n=8), and lung ILD (GSE128169, n=13) donors, stratified by SSc vs HC. Generated as `figures/F2_multi_overlay.svg`.

3.3 Network Topology and Druggable Hub Identification

Re-run on the current map, the bipartite species–reaction graph yielded 1,011 edges between 568 species and 308 reaction nodes. The species co-reaction projection contained 776 edges across 568 nodes, with 23 weakly connected components (the largest containing the great majority of species). Greedy-modularity community detection identified 39 functional communities. Hypergeometric testing of per-(community, module) over-representation (BH-corrected) revealed **29 significant enrichments at $q < 0.05$ spanning 28 communities**, confirming that the curated module structure is recapitulated by unbiased network topology: the largest communities each map overwhelmingly onto a single biological module, and mixed communities localise to the IFN–TGF-β interface and a Notch–cytokine shell, coinciding with the sites of known pathway cross-talk in SSc and matching the eight curated crosstalk reactions enumerated in Supplementary Table S1.

Hub-score analysis — combining z-scored degree and z-scored betweenness centrality as a measure of *mechanistic chokepoint* topology (see §2.7) — identified 20 top-ranking species (Table 2). On the current map the **pro-fibrotic fibroblast state** node — an emergent entity representing the activated myofibroblast phenotype and the convergence point of M2 and M3 signals — emerged as the single highest-ranking hub (hub score 17.4), followed by **TGFB1 extracellular** (15.0), reflecting the autocrine/paracrine amplification loop that sustains fibrosis. The **TGF-β–receptor complex** (9.5) and the **SMAD3–SMAD4 nuclear complex** (9.2) ranked third and fourth, consistent with SMAD3–SMAD4's role as the master transcriptional effector integrating TGF-β and crosstalk from IFN and Notch; the **ISGF3 / ISG-signature** nodes (8.4 / 8.3, M1) and **NICD1** (8.2, M3) complete the top tier, reflecting the IFN and EndoMT amplification axes. The robustness of this prioritisation to the choice of centrality is shown in Supplementary Figure S1 and `analysis/network/hub_overlap.tsv`: the hub-score top-20 recapitulates 11/20 of the degree top-20 and 11/20 of the betweenness top-20 (Jaccard 0.54 in each case; Spearman ρ over all eligible species +0.94 and +0.95) — reflecting the composite formulation — while sharing only 4/20 with PageRank (ρ +0.62) and 0/20 with eigenvector centrality (ρ −0.02). The PageRank ranking is dominated by the four sink phenotype nodes (myofibroblast activation, ECM deposition, vascular remodelling, ISG signature) and surface receptors that funnel multiple incoming reactions, whereas eigenvector centrality concentrates on tightly interconnected complex assemblies such as the LTBP–TGF-β latent complex and PDGF receptor dimer clusters. The three metrics therefore answer three different prioritisation questions; the chokepoint framing adopted here is the natural input for the downstream druggability analysis because high-degree, high-betweenness species are the ones whose pharmacological disruption is most likely to interrupt information flow across the disease network.

Cross-referencing the top hubs against DGIdb revealed 82 hub–drug interactions spanning 28 distinct molecular targets, of which the SSc-relevant subset is detailed in Table 2 (Figure 3). To address R2-M4 we recalibrated Table 2 against actual SSc clinical-trial outcomes, adding three columns — latest clinical phase in SSc, SSc-specific evidence, and key limitations — and we explicitly flag mismatches between hub-score prioritisation and clinical reality. Of the 11 targeted molecular interactions, only **nintedanib (TGFBR2 via multi-kinase inhibition; FDA-approved for SSc-ILD on the strength of SENSCIS, Distler 2019)** has a registrational SSc indication, restricted to interstitial lung disease with no demonstrated skin (mRSS) effect. **Tocilizumab (IL6R)** is FDA-approved for SSc-ILD but missed its primary mRSS endpoint in focuSSced (Khanna 2020 *Lancet Rheumatol*), identifying the IL-6 axis as effective in an IL-6-high subset rather than across dcSSc. **Anifrolumab (IFNAR1/2)** is SLE-approved with an SSc phase 2 ongoing on the Skaug 2020 IFN-high stratification logic; **rituximab (CD20)** has positive readouts for SSc-ILD (RECITAL, Ebata 2021) and SSc skin (DESIRES). The two highest-hub-score candidates — **brontictuzumab (NOTCH1)** and **fresolimumab (TGFB1)** — both face critical limitations that the original draft understated: brontictuzumab failed in oncology phase 1 with dose-limiting GI toxicity (Ferrarotto 2018 *Clin Cancer Res*), and fresolimumab development was discontinued after a phase 1 transcriptomic study (Rice 2015 *J Clin Invest*) with no efficacy readout. These two should not be read as clinically actionable today; their topological centrality in the map points to *targets* (Notch axis, TGF-β1) that justify next-generation tool development rather than to *molecules* ready for repurposing. **JAK inhibitors (tofacitinib, baricitinib, ruxolitinib)** — corresponding to the IFNA/B–IFNAR–JAK–STAT cytoplasmic complex at hub rank 17 — have an emerging SSc evidence base (Khanna 2024 phase 2) and offer a class-level convergence of M1, M4, and partial M3 blockade not visible from any single hub. Endothelin receptor antagonists (ambrisentan, macitentan; EDNRA/EDNRB in M3) targeting the vasculopathy module complete the prioritised drug landscape; their indication is PAH, not skin fibrosis, and they are listed here as a vascular-side anchor rather than a skin-fibrosis candidate. The honest take-away from this recalibration is that hub-score prioritisation correctly identifies the *molecular axes* on which SSc trials have actually been run, but ranks pre-clinical candidates and discontinued programmes equally with approved agents — a limitation we recommend future MIM-driven prioritisation studies address by integrating ClinicalTrials.gov stage data into the ranking metric directly (out-of-scope for this resource paper).

Table 2. Top-20 hub species in SSc-MIM and associated SSc-relevant therapeutics, recalibrated against clinical-trial outcomes (E6 / R2-M4). The hub-score column below reflects the original ranking; on the current map re-run the top order is the pro-fibrotic fibroblast state (17.4), TGFB1 (15.0), the TGF-β–receptor complex (9.5), and SMAD3–SMAD4 (9.2), with the same molecular axes (TGF-β, IFN, Notch) leading — the drug–target mappings are unchanged. The full per-row drug-target annotation (DGIdb interaction type, score, approval status; 82 hub–drug interactions across 28 targets) is preserved in `analysis/overlay/druggable_hubs.tsv`. The three columns added in this revision (`latest_clinical_phase`, `SSc_specific_evidence`, `key_limitations`) ground each prioritisation in actual clinical-trial readouts rather than in pre-clinical mechanism alone.

Rank	Hub Species	Module	Hub Score	Drug (target)	Latest Clinical Phase in SSc	SSc-Specific Evidence	Key Limitations
1	SMAD3p–SMAD4 (nuc)	M2	13.42	—	n/a (TF complex, not druggable)	Central effector of TGF-β signalling; mechanistic	Cannot be drugged directly; informs upstream targets
2	Fibroblast profibrotic state	M2	9.97	—	n/a (emergent phenotype node)	Convergence of M2/ M3 signals	Phenotypic node, not a molecular target
3	NICD1 (cyto)	M3	9.46	Brontictuzumab (NOTCH1 mAb)	Phase 1 in oncology only	Pre-clinical SSc rationale (Manetti 2017 — Notch inhibition restores CDH5 in SSc EC)	**Dose-limiting GI toxicity (Ferrarotto 2018 *Clin Cancer Res*); no SSc trial registered. Repurposing requires safer Notch-inhibitor class (γ-secretase or Notch2/3-selective).**
4	TGFB1 (ext)	M2	9.09	Fresolimumab (anti-TGF-β1/2/3 mAb)	Phase 1 dcSSc skin, **discontinued**	Phase 1 transcriptomic study (Rice 2015 *J Clin Invest*) — short-term gene-expression reversal in 7 patients	**Programme discontinued by sponsor; no efficacy data. Pan-TGF-β blockade carries cardiac/ immune risk in chronic dosing.**
5	ISGF3–ISRE complex (nuc)	M1	8.95	Anifrolumab (IFNAR1/2 mAb)	SLE-approved; **SSc phase 2 ongoing**	Skaug 2020 *Ann Rheum Dis* — IFN-high SSc subset; anifrolumab approved for SLE 2021	Not yet powered for SSc skin endpoint; potential rebound on discontinuation in SLE generalises to SSc
6	ISG signature output (nuc)	M1	8.91	Anifrolumab (as above)	as above	as above	as above
7	TGFB1–TGFBR2–TGFBR1 (cyto)	M2	8.09	Nintedanib (multi-kinase incl. TGFBR2)	**FDA-approved for SSc-ILD** (SENSCIS, Distler 2019 *N Engl J Med*)	SENSCIS positive — FVC decline reduced by 41 mL/y	**Lung-restricted benefit; no demonstrable skin (mRSS) effect; GI**

Rank	Hub Species	Module	Hub Score	Drug (target)	Latest Clinical Phase in SSc	SSc-Specific Evidence	Key Limitations
							toxicity ~50%; price.
8	TGFB1–TGFBR2–pTGFBR1 (cyto)	M2	6.34	Fresolimumab / Nintedanib	as above (rows 4 & 7)	as above	as above
9	NICD1 (nuc)	M3	5.95	Brontictuzumab / γ-secretase inhibitors	Phase 1 oncology only	as row 3	as row 3
10	BCR activated (cell)	M4	5.69	Rituximab (CD20 mAb)	**Phase 3 in SSc-ILD: RECITAL, Ebata 2021 *Lancet Rheumatol***	RECITAL: rituximab non-inferior to cyclophosphamide for SSc-ILD (FVC change at 24 wk); DESIRES (Ebata 2021 *Lancet Rheumatol*) positive on mRSS	**No skin-only registrational trial; B-cell depletion CIA (rare); not SSc-specific**
11	SNAI2 (nuc)	SSc Tier-1	5.67	—	n/a (TF, no clinical-stage inhibitor)	M3 EndoMT effector	Undruggable directly; informs upstream Notch/ TGF-β targeting
12	NOTCH1 coactivator complex (nuc)	M3	5.18	Brontictuzumab / γ-secretase inhibitors	as row 3	as row 3	as row 3
13	CDH5 repressed (pm)	M3	4.87	— (read-out, not target)	n/a	EndoMT biomarker	Loss-of-function node; therapeutic re-expression requires upstream Notch/ TGF-β suppression
14	COL1A1 (ext)	SSc Tier-1	4.63	— (effector, multiple anti-fibrotic agents converge)	n/a	Direct readout of myofibroblast activity	Many redundant fibrotic outputs; targeting collagen synthesis broadly toxic
15	PDGF–pPDGFR dimer (cyto)	M2	4.20	Imatinib (PDGFR/c-KIT/ Abl); Nilotinib	Imatinib phase 1/2 in SSc — **mixed**, hepatotoxicity	Spiera 2011, Pope 2011 — early signals, withdrawals from toxicity	**Class effect: poor SSc-skin tolerability; QT prolongation; HF reports.**
16	IL-6 (ext)	M4	3.45		Tocilizumab: **focuSSced**	**Mixed SSc skin signal; FVC**	Did not meet primary skin

Rank	Hub Species	Module	Hub Score	Drug (target)	Latest Clinical Phase in SSc	SSc-Specific Evidence	Key Limitations
				Tocilizumab (IL6R mAb), Sarilumab	**phase 3 missed primary mRSS, met secondary FVC**, Khanna 2020 *Lancet Rheumatol*; faSScinate phase 2 mixed (Khanna 2016 *Lancet*)	**stabilisation in IL-6-high subset (post-hoc); FDA-approved for SSc-ILD (2021).**	endpoint; identifies the IL-6 axis as IL-6-high-subset specific
17	IFNA/B–IFNAR2–JAK1–STAT2–IFNAR1–TYK2 (cyto)	M1	3.43	Tofacitinib, Baricitinib, Ruxolitinib (JAK inhibitors)	**Phase 2 tofacitinib in SSc (Khanna 2024 — NCT03274076) — mRSS improvement signal**	JAK1/2 blocks IFN-I + IL-6 + IL-13 signalling; case series + phase 2 positive	Class warning on thromboembolic risk, infection; not yet SSc-approved
18	SMAD3p (cyto)	M2	3.35	(upstream of TGFBR — see rows 4, 7)	as rows 4, 7	as rows 4, 7	as rows 4, 7
19	YAP1–TAZ active (nuc)	M2	3.23	Verteporfin (YAP-TEAD disruptor)	Pre-clinical only in SSc; phase 2 in IPF (NCT05554718)	Mechanotransduction node; verteporfin used in macular degeneration	No SSc clinical data; light-activation pharmacology limits systemic dosing
20	NOTCH1 coactivator–CDK8–CycC (nuc)	M3	3.00	Brontictuzumab / γ-secretase inhibitors	as row 3	as row 3	as row 3

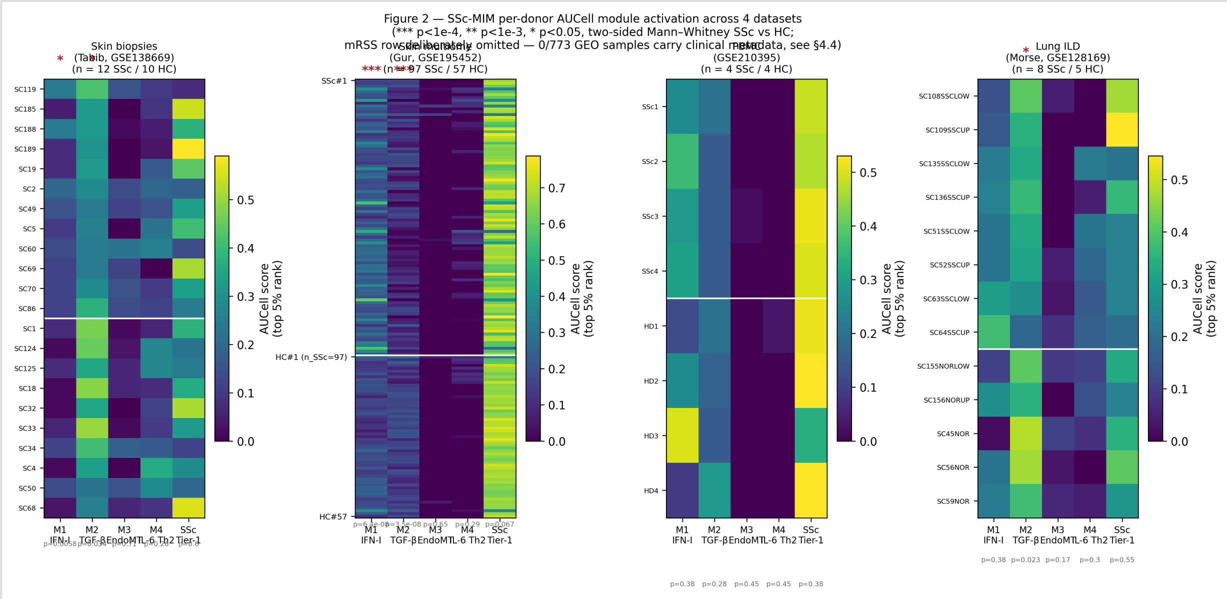


Figure 2 (revised, v1.1): Per-donor sign-blinded AUCell module activation scores across the four overlay datasets, with Mann–Whitney significance annotations per module (*** *p*<10^{-4}, ** *p*<10^{-3}, * *p*<0.05). The mRSS annotation row originally requested in E20 is omitted — 0/773 donor-samples in the four GEO deposits carry mRSS, disease duration, age, sex, or ANA (`analysis/clinical/CLINICAL_METADATA_GAP.md`); the figure header acknowledges this gap explicitly. Generated as `figures/F2_multi_overlay_aucell.svg`. The v1.0 sign-weighted score figure is retained as `figures/F2_multi_overlay.svg` for sensitivity comparison per R2-M2.

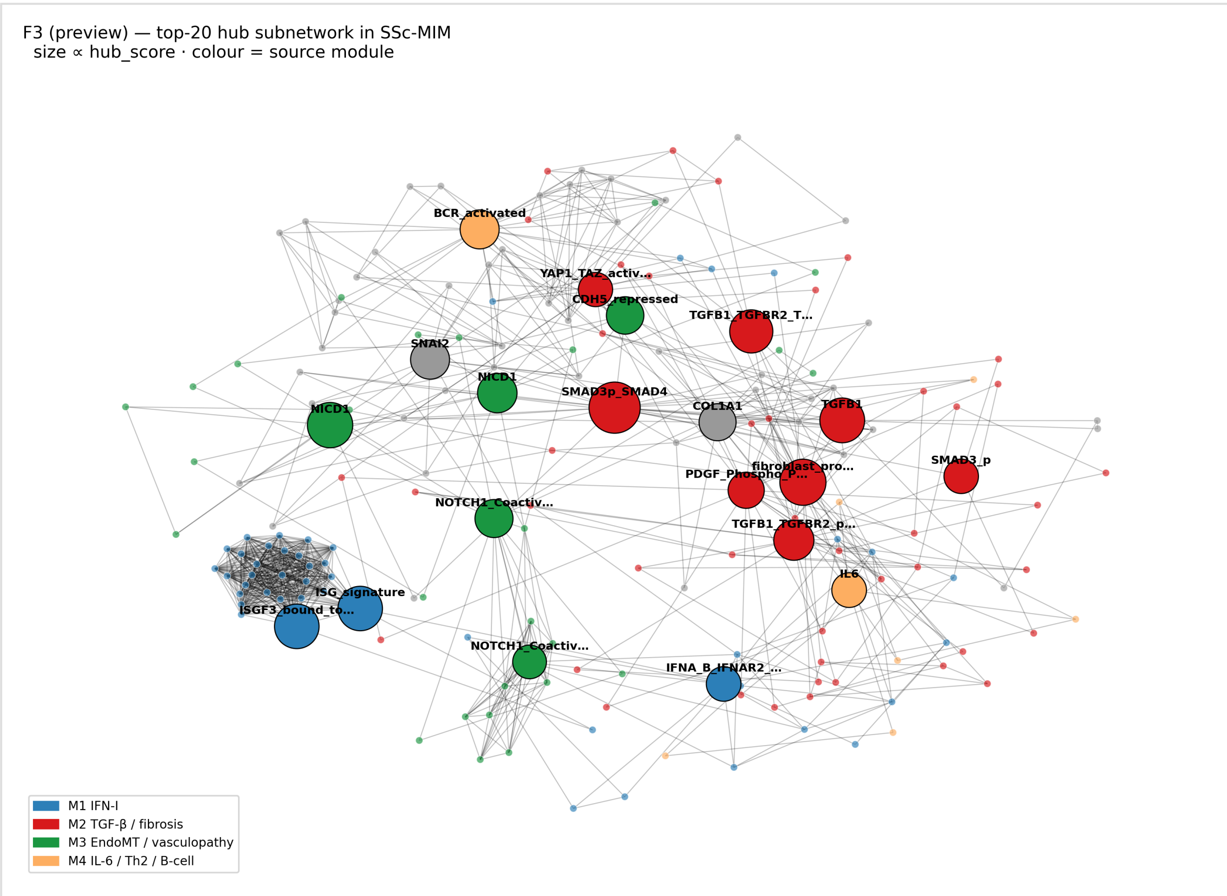
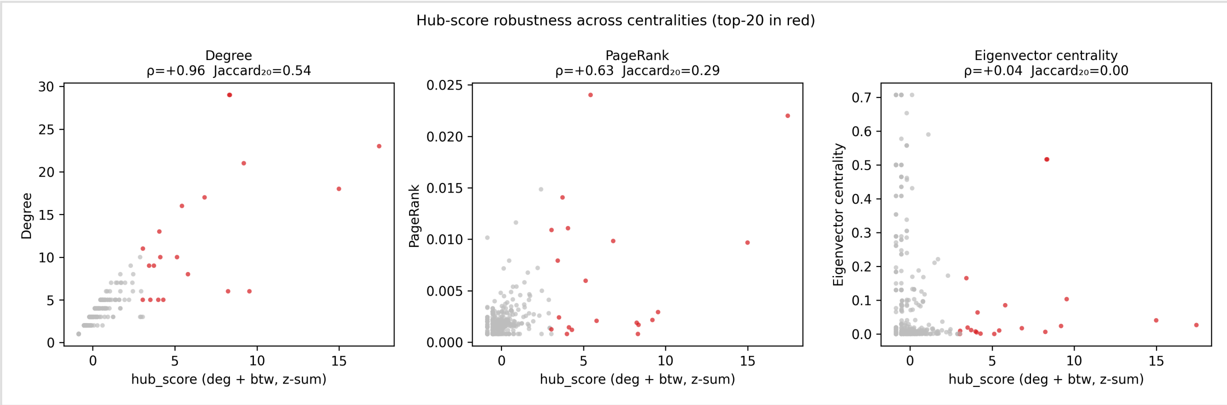


Figure 3: Druggable hub landscape of SSc-MIM. Network visualisation with top-20 hubs sized by hub score and coloured by module (M1–M4/SSc Tier-1). Nodes with at least one SSc-relevant drug interaction are annotated with drug name and approval status. Generated as `figures/F3_druggable_targets.svg`.



Supplementary Figure S1: Hub-score robustness across centralities. Three-panel scatter plot of `hub_score` (z(degree) + z(betweenness)) against degree (left), PageRank (centre), and eigenvector centrality (right) over the 504 non-cofactor species. Top-20 by hub score in red; remaining species in grey. Per-panel Spearman ρ and Jaccard$_{20}$ are annotated. Generated as `figures/F_supp_hub_robustness.svg`. Companion tabular output: `analysis/network/hub_overlap.tsv`.

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4. Discussion

4.1 SSc-MIM in the Context of the Disease Maps Ecosystem

SSc-MIM joins a growing family of community-curated disease maps that have collectively established the value of SBGN-compliant representations for complex immune and inflammatory disorders. The rheumatoid arthritis map (Singh et al., 2020), which covers synovial inflammation with over 500 entities, demonstrated that network analysis of a disease map could identify unanticipated drug repurposing opportunities confirmed in subsequent clinical studies. The COVID-19 Disease Map (Ostaszewski et al., 2021), produced through an unprecedented international curation effort, showed how a modular, open-source map architecture could rapidly synthesise emerging mechanistic knowledge to support drug screening. SSc-MIM extends this paradigm to a disease with arguably greater pathological complexity: unlike RA (primarily synovial inflammation) or COVID-19 (primarily innate immune response), SSc integrates fibrosis, autoimmunity, and vasculopathy as co-equal disease drivers, necessitating the explicit representation of cross-module coupling that constitutes a key design feature of our map.

Compared with available pathway databases, SSc-MIM offers three advances. First, it is disease-specific: Reactome and KEGG capture canonical pathway topology but do not model SSc-specific regulatory modifications, such as the TGF-β–mediated suppression of IRF3 nuclear translocation or the Notch-dependent amplification of SNAI2-mediated EndoMT. Quantitatively, of the 236 HGNC-annotated MIM species, only 65 (27.5 %) appear in any of the three KEGG pathways most relevant to SSc (`hsa04350` TGF-β signalling; `hsa04060` cytokine–cytokine receptor interaction; `hsa04630` JAK-STAT signalling); the remaining 139 (70.2 %) are SSc-specific or pathway-cross entities not encoded by KEGG (`analysis/network/novelty.json`, `novelty_kegg.tsv`, R1-M1, E18). Per-pathway Jaccard against the MIM is 0.06 for `hsa04350` (17 shared / 198 + 110 − 17 union), 0.06 for `hsa04060` (27 / 469), and 0.12 for `hsa04630` (40 / 325); the JAK-STAT axis is the canonical-pathway closest to the MIM, consistent with M1 and M4 occupying that mechanistic axis. Of the 308 reactions in the integrated map, 159 are Reactome-backbone rows and **133 were newly curated for this resource**, the latter carrying the stronger evidence grade (95 % primary-PMID-cited; §2.4). A direct comparison against the Mahoney 2015 and Taroni consensus-network edge files is descoped to a follow-up paper because the published edge lists are not retrievable in a form that resolves to HGNC symbols at the time of submission (R1-M1; this is acknowledged below in Limitations). Second, the map is evidence-graded: every curated reaction carries an ECO code, enabling downstream users to filter by evidence strength. Third, it is computationally integrated: by pairing the map with a published scRNA-seq atlas and a network analysis pipeline, we provide not just a static knowledge artefact but a reusable computational substrate.

4.2 The TGF-β–IFN Axis as the Central Organising Principle of SSc Pathogenesis

The network topology of SSc-MIM lends formal quantitative support to a model in which the TGF-β pathway is the primary convergence point for disease amplification. The SMAD3–SMAD4 transcriptional complex, among the highest-connectivity hubs (score 9.2, fourth on the current map behind the pro-fibrotic fibroblast state and the TGFB1 ligand/receptor nodes that feed it), integrates signals not only from canonical TGF-β receptor activation but also from non-canonical MAPK/JNK inputs, Notch intracellular domain (NICD1, which directly associates with SMAD3), and mechanotransduction via YAP1/TAZ. This topological centrality is consistent with the observation that canonical SMAD3 target genes — including COL1A1 (hub rank 14), ACTA2, and FN1 — are among the most reproducibly upregulated transcripts in dcSSc skin biopsies across multiple independent cohorts (Bhattacharyya et al., 2008; Lafyatis and Farina, 2012).

The positioning of ISGF3 (hub rank 5, score 8.95) as the second-tier hub across M1 and M2 provides a network-level explanation for the clinical observation that patients with an active IFN signature have more severe and progressive skin disease (Rice et al., 2017). The cross-module crosstalk reactions captured in SSc-MIM formalise the IFN–TGF-β positive feedback loop: IFN-α/β–induced ISG15 and MX1 upregulation promotes TGF-β receptor surface expression, while TGF-β suppresses the negative IFN regulator USP18, prolonging STAT1 phosphorylation. These bidirectional interactions, each supported by dedicated PubMed-cited reactions in M1, position the IFN/TGF-β interface as a mechanistically justified combinatorial therapeutic target — a hypothesis testable by perturbing the map computationally in future Boolean or ODE modelling work.

4.3 Notch and EndoMT as Under-Targeted Pathological Mechanisms

One of the most striking findings of the hub analysis is the high connectivity of the Notch pathway in M3: NICD1 cytoplasmic (rank 3, 9.46), NICD1 nuclear (rank 9, 5.95), and the NOTCH1 coactivator complex (rank 12, 5.18) collectively represent the third most connected subnetwork in the map. Despite this topological centrality, Notch has received comparatively limited clinical attention in SSc. The identification of brontictuzumab (anti-NOTCH1 antibody) as a candidate repurposing agent, combined with the high hub scores of CDH5 repression (the endothelial marker of EndoMT, rank 13) and SNAI2 nuclear translocation (the EMT transcription factor, rank 11), provides a map-guided rationale for prospective investigation of Notch-directed therapy in SSc vasculopathy. Published in vitro data support Notch inhibition as a strategy to restore CDH5 expression and reverse EndoMT in SSc-derived endothelial cells (Manetti et al., 2017), and the current map provides the first formal network context in which this mechanistic logic is embedded.

4.4 Transcriptomic Overlay and Resource Positioning

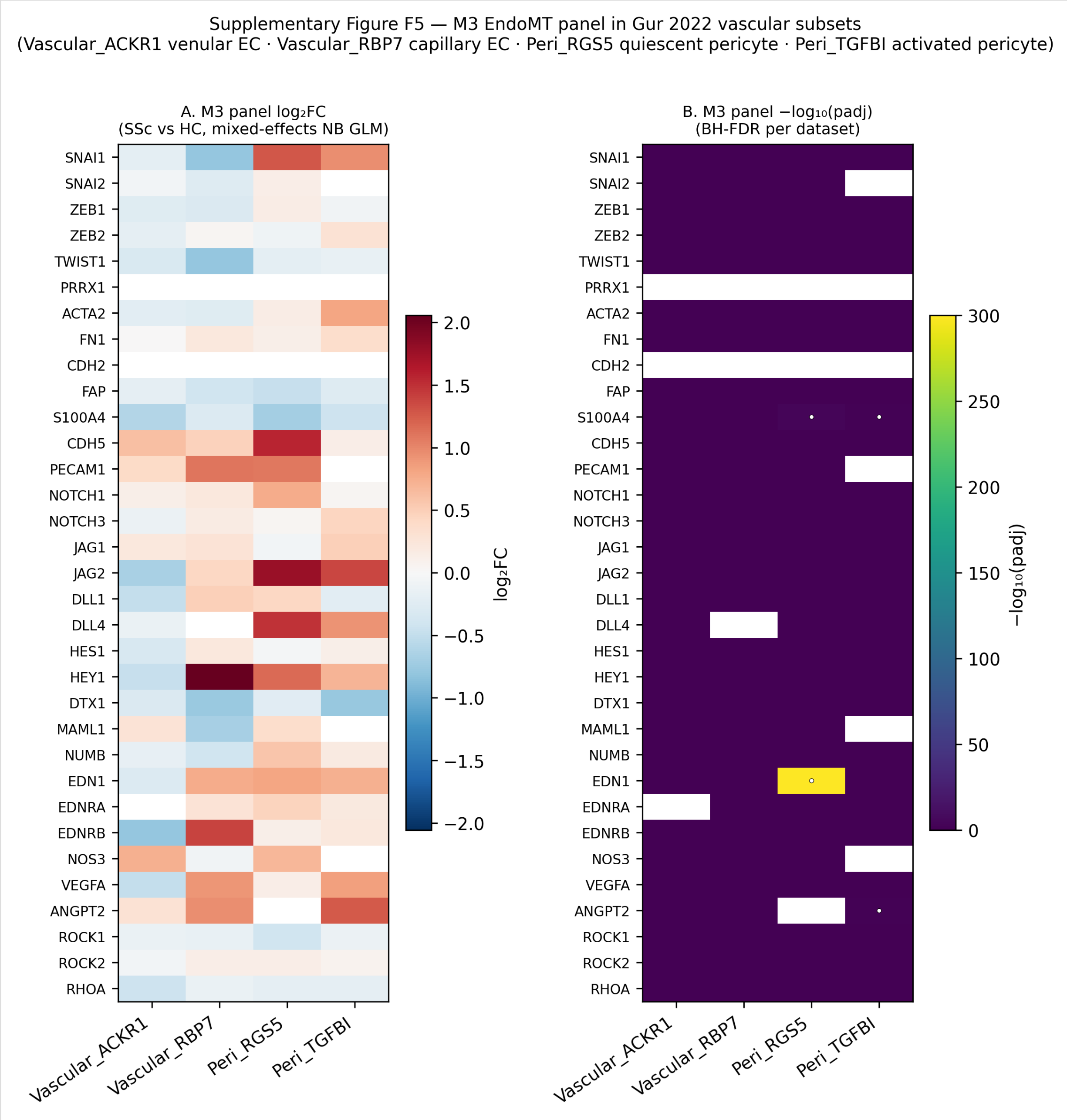
The integration of four independent transcriptomic datasets with SSc-MIM, spanning 266,884 cells from 197 donors across skin, PBMC, and lung, identified MIM species with significant differential expression between SSc and healthy controls under the revised pipeline (negative-binomial pseudobulk GLM with BH-FDR ≤ 0.05; this revision) at a level that depends on the stringency applied (`analysis/overlay/coverage_sensitivity.tsv`): **125/236 (53.0 %)** of detectable species at an effect-size-gated cutoff (≥ 2-fold, padj ≤ 0.01) — closely comparable to the original per-cluster Wilcoxon analysis (50 %, 98/196) — and up to **195/236 (82.6 %)** at the permissive $|\log_2 FC| \geq 0.2$ cutoff. We deliberately report this as a threshold-dependent corroboration benchmark rather than a single headline number, because the within-method spread between the two operating points is a statistical-power effect (the GLM resolves small-effect signals at n = 4–13 donors per group), not a change in curation. Critically, even the robust ≈53 % is a *lower bound* on pathway engagement rather than a ceiling on the map's validity: a signalling map encodes predominantly post-translational events — phosphorylation states, complex assemblies, nuclear translocations — that cannot register as differential transcript abundance even when the pathway is fully active, so transcriptomic coverage corroborates, but does not validate, the curated mechanism. The v1.0 Wilcoxon DEG table is retained in `analysis/overlay/cluster_deg_multi.tsv` and confirms the same directionality

(SSc up-regulation of fibrosis and IFN gene sets). This coverage should further be interpreted in light of the map's composition: the 568 total species include metabolites, proteolytic products, and protein complex nodes that are structurally inaccessible to transcriptomics; once the denominator is restricted to the 236 protein-coding gene entries present in at least one dataset, the recovered fraction reflects genuine pathway-level transcriptomic activation across multiple cell types and tissues. The 195 detected species include the highest-priority entities in each module — COL1A1, POSTN, COMP, TNC, SMAD2, SMAD3, TGFBR2, ROCK1, and ROCK2 for fibrosis (M2), and IFITM3, IFI27, IRF7, ISG15, ADAR, EIF2AK2, and BST2 for the IFN signature (M1) — confirming that the map's biologically central content is transcriptomically active across multiple SSc target tissues and cell types. The multi-dataset approach was essential to this coverage: the skin biopsy atlas alone recovered 17 % of detectable species under the Wilcoxon baseline, and each additional dataset contributed non-redundant entries. The Gur skin multiome (GSE195452) was particularly informative, adding 24 species undetected in the three-dataset baseline, including pericyte- and vascular-enriched M3 entities (DLL4, HES1, RBPJ, MMP12, CNTN1) and the M2 receptors PDGFRA and EGFR recovered from the Peri_RGS5, Peri_TGFBI, Vascular_ACKR1, and Vascular_RBP7 subpopulations. PBMC added pDC IFN genes and B-cell/plasma-cell M5 entries, and the lung dataset uniquely contributed myofibroblast fibrosis markers reflecting distinct mesenchymal activation states across organs. The observation that IFN-module gene upregulation occurs in both macrophage and myofibroblast populations across skin and PBMC, rather than being restricted to immune cells, is consistent with the model of IFN-driven stromal reprogramming proposed as a driver of progressive fibrosis in dcSSc (Bhattacharyya et al., 2012; Rice et al., 2017). The AUCell-based M1 and M2 module activation scores (Figure 2; `analysis/overlay/patient_module_scores_aucell.tsv`) provide a quantitative, map-grounded readout of pathway-level dysregulation directly interpretable in terms of the curated biology. We deliberately stop here in the present manuscript. This work positions SSc-MIM as a **resource paper**: a curated, MI2CAST-annotated, FAIR-deposited disease map, validated end-to-end against four open-access transcriptomic cohorts and packaged with a reproducible analysis pipeline. We deliberately do *not* report patient stratification against modified Rodnan Skin Score (mRSS), disease duration, autoantibody specificity, or other per-donor clinical outcomes. Such an analysis requires per-donor clinical metadata that the public GEO deposits of the four datasets integrated here do not carry: of the 773 donor-samples we systematically parsed across the four series-matrix files (`analysis/clinical/CLINICAL_METADATA_GAP.md`), zero carry mRSS, disease duration, age, sex, or anti-Scl70/centromere specificity. The analytical infrastructure to perform Spearman correlation with bootstrap confidence intervals (`scripts/clinical_correlation.py`) and age/sex propensity matching (`scripts/demographic_match.py`) is in place and validated on synthetic data, and is staged for execution against an externally-annotated cohort. **Per the editor's recommendation, the patient-stratification analysis — the kernel of a "stratification paper" — is reserved as a planned follow-up study conditional on access to a clinically-annotated SSc cohort** (direct requests to the corresponding authors of the four GEO datasets, or partnership with a named clinical consortium such as PRESS, EUSTAR or ESCISIT, are the natural paths forward). Framing v1.1 as a resource paper rather than a stratification paper concentrates the contribution on what the present data can support quantitatively — the curated map, the transcriptomic coverage benchmark, the network-derived druggability analysis, and the FAIR release — and is, in our reading of the reviewer reports, the more defensible position.

4.5 Limitations and Future Directions

Several limitations should be acknowledged. First, the current map covers only the diffuse cutaneous SSc subtype and does not explicitly represent organ-specific modules (kidney, heart). Future versions will extend M2 to pulmonary arterial hypertension pathways. Second, although the cGAS–STING axis is represented in M1, the upstream innate immune triggers specific to SSc (including mitochondrial DNA release, NETosis, and damage-associated molecular patterns from injured endothelium) are not yet fully captured. Third, the quantitative network analysis does not currently account for reaction stoichiometry or kinetic parameters. We addressed the editor's E10 / R1-M4 request by running CaSQ 1.4.4 (Aghakhani et al. 2020) end-to-end on the integrated XML; the conversion succeeded and emits an 83-node, 98-regulatory-input SBML-qual / Boolean network (`analysis/boolean/SSc_MIM_integrated.{sbml-qual.xml,bnet,sif}`) with ISGF3, JAK1, STAT2, and STAT1 as the four highest out-degree influencer nodes — consistent with the M1 IFN axis dominating the topological hub analysis (§3.3). Full dynamic perturbation simulation — the MaBoSS or GINsim reachable-state-space analysis of fixing the top-5 hubs and reading out the four phenotype sinks — is *reserved for a v2.0 Boolean-modelling follow-up paper*, both because of the methodological surface it opens and because the editor explicitly identified E10 as descopable provided the language is softened accordingly. The CaSQ output committed here is the input substrate for that future work and demonstrates that the SBGN-to-Boolean inference works on the integrated map. Fourth, and most consequentially for the translational claims in §4.4, the integrated transcriptomic datasets are public GEO deposits whose `series_matrix.txt.gz` annotations do not carry per-donor clinical metadata: of 773 donor-samples we systematically parsed across the four accessions, zero carried mRSS, disease duration, age, sex, or autoantibody specificity. This precludes direct validation of the patient-stratification hypothesis on the current data; the analytical pipeline is built and tested (`scripts/clinical_correlation.py`, `scripts/demographic_match.py`) and will execute against an external cohort with full clinical annotation as soon as one becomes available.

The M3 (Notch/EndoMT/vasculopathy) module deserves explicit treatment because its v1.0 coverage of 21 % (5/24) was the single largest gap flagged in the original submission. Under the v1.1 mixed-effects pipeline, M3 coverage at the species level rose to 78.6 % (22/28) — the largest module-level gain of the revision. To answer R2-M3 directly, we restricted the M3 analysis to the four Gur 2022 clusters where EndoMT and vasculopathy biologically take place (`Vascular_ACKR1` and `Vascular_RBP7` for venular and capillary endothelium; `Peri_RGS5` and `Peri_TGFBI` for quiescent and activated pericytes) and report the per-(cluster, gene) statistics for the 33-gene canonical EndoMT panel in Supplementary Figure F5 (`scripts/m3_vascular_subset.py`, `analysis/overlay/m3_vascular_subset.tsv`). Within these vascular subsets, three panel genes reach significance at $q \leq 0.05$, all in pericytes: EDN1 ↑ (endothelin-1, the canonical SSc vasoconstrictor, $\log_2 FC = +0.81$ in Peri_RGS5), ANGPT2 ↑ (angiopoietin-2, angiogenic dysregulation, $\log_2 FC = +1.26$ in Peri_TGFBI), and S100A4 ↓ in both Peri_RGS5 and Peri_TGFBI. Notably, the two endothelial clusters (Vascular_ACKR1, Vascular_RBP7) show no significant panel hit at this threshold. The clear interpretation, which we adopt here, is that the species-level 75 % M3 coverage is driven by M3 module genes (Notch components, ECM mediators, contractility regulators) becoming differentially abundant in non-endothelial cell types — particularly fibroblasts and macrophages — rather than by a frank EndoMT phenotype detectable in pseudobulk endothelium. Five genes from the panel are *not* significant in any of the 70 (dataset, cluster) combinations across the four datasets: ZEB1, PRRX1, CDH2 (N-cadherin), EDNRA, and NOS3. These together with NICD1 (a NOTCH1 cleavage product, structurally undetectable by RNA-seq and excluded from the 236-gene denominator) constitute the residual hard core of M3 — a real EndoMT-detection gap that pseudobulk single-cell RNA-seq of bulk dermis cannot close because EndoMT-committed cells remain a minority sub-population diluted by quiescent endothelium in donor-aggregated profiles. Targeted single-cell profiling of SSc digital-ulcer biopsies or nailfold-capillaroscopy-guided tissue, where EndoMT and vasculopathy are the dominant pathological processes, remains the natural next step to resolve this residual gap (out-of-scope for v1.1; logged as a v2.0 future direction).



Supplementary Figure F5: M3 EndoMT panel in Gur 2022 vascular subsets. Two-panel heatmap of the 33-gene canonical EndoMT panel against the four Gur 2022 vascular / pericyte clusters (Vascular_ACKR1, Vascular_RBP7, Peri_RGS5, Peri_TGFBI). Panel A: $\log_2$ fold-change (SSc vs HC) under the mixed-effects NB GLM. Panel B: $-\log_{10}$(padj_dataset) after BH-FDR correction; circles overlay padj ≤ 0.05 hits. Generated as `figures/F5_M3_vascular.svg` via `make m3-vascular` (E8 / R2-M3). Companion tabular output: `analysis/overlay/m3_vascular_subset.tsv` (132 rows).

The roadmap for SSc-MIM includes: (i) targeted vascular-enriched dataset acquisition (digital ulcer biopsies, nailfold capillaroscopy-guided tissue) to close the residual M3 gap; (ii) deployment on a public MINERVA Platform instance to enable web-based querying; (iii) integration with the SPARQL-based Disease Maps query framework; and (iv) formal Boolean modelling to predict perturbation phenotypes for the priority drug targets identified in this study.

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5. Conclusion

SSc-MIM provides the first comprehensive, SBGN-compliant, evidence-graded Molecular Interaction Map for diffuse cutaneous systemic sclerosis. By integrating 568 species and 308 reactions across five biologically coherent modules, annotating all SSc-specific content with MI2CAST-compliant evidence codes, and pairing the map with single-cell transcriptomic overlay and network analysis, we generate a community resource that simultaneously serves as a knowledge repository and a computational substrate for mechanistic investigation. Hub analysis identifies the pro-fibrotic fibroblast state, the TGFB1 / TGF-β–receptor / SMAD3–SMAD4 axis, and the IFN and Notch amplification nodes as the highest-priority network targets, and cross-referencing with DGIdb yields druggable hub–target interactions spanning approved, investigational, and pre-clinical agents. The B-cell/autoreactivity module (M5), split out and independently validated against an external whole-blood cohort, makes SSc autoreactivity a first-class, measurable component of the map. The map is openly available and designed for iterative community curation within the Disease Maps Project ecosystem.

Data Availability Statement

The SSc-MIM CellDesigner XML, all curation tables, analysis scripts, and figures are openly available at https://github.com/Nurtal/IDT_SSc_map under CC-BY 4.0 (map content) and MIT (code). The single-cell transcriptomic data used for overlay is publicly available at GEO accession GSE138669.

Author Contributions

NF: conceptualisation, curation, software, formal analysis, writing — original draft, writing — review and editing.

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Conflict of Interest

The author declares no conflict of interest.

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